Project Milestone 4: Interpretability and Failure Analysis for Video Moment Retrieval

Temporal Sentence Grounding in Videos

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Abstract

Video moment retrieval (VMR) requires localizing temporal segments in untrimmed videos given natural language queries. While recent DETR-style transformer models achieve strong performance, their internal mechanisms and failure modes remain poorly understood. We reproduce three baselines—Moment-DETR, QD-DETR, and CG-DETR—on the QVHighlights benchmark and conduct three interpretability analyses. First, we perform temporal bias analysis, showing that models achieve 7× above trivial baselines but exhibit a dominant failure mode on short moments (<10s), where R1@0.5 drops by roughly 12–16 points due to systematic length overprediction. Second, we conduct verb vs. noun sensitivity ablation by zeroing POS-tagged token features, revealing that object nouns are 2.6× more important than action verbs for grounding in QD-DETR. Third, we perform individual attention head ablation across all 32 heads, finding that no single head is specialized for boundary prediction—temporal grounding emerges as a distributed property of the full transformer. These findings identify concrete failure modes and suggest that improving short-moment handling and action-verb representations are the most promising directions for future work.

1 Introduction

Video moment retrieval (VMR), also known as temporal sentence grounding in videos (TSGV), is the task of localizing a temporal segment $[t_s, t_e]$ within an untrimmed video that semantically corresponds to a given natural language query [Gao et al., 2017]. For example, given the query "the person adds salt to the pan" and a 150-second cooking video, a VMR system must predict the precise start and end timestamps of the matching moment.

This task is fundamental to video understanding and has applications in video search, surveillance, content creation, and assistive technologies. Unlike video classification or action detection, VMR requires jointly understanding visual content and natural language, then performing fine-grained cross-modal alignment to identify temporal boundaries.

Recent approaches based on the Detection Transformer (DETR) [Carion et al., 2020] architecture have achieved state-of-the-art results by framing moment retrieval as a set prediction problem. Moment-DETR [Lei et al., 2021] introduced this paradigm, while subsequent models like QD-DETR [Moon et al., 2023b] and CG-DETR [Moon et al., 2023a] improved upon it by conditioning video representations on the query and calibrating cross-modal attention, respectively.

Despite their strong aggregate performance, several questions remain open about these models:

1. Do models learn genuine cross-modal grounding, or do they exploit temporal biases in the dataset (e.g., predicting near the video center)?
2. Which linguistic components—action verbs or object nouns—drive the model's grounding decisions?
3. Are there specialized attention heads responsible for temporal boundary prediction, or is this ability distributed across the architecture?

We address these questions through three complementary analyses on the QVHighlights benchmark [Lei et al., 2021], using the Lighthouse framework [Nishimura et al., 2024] for reproducible evaluation.

Our contributions are:

- **Temporal bias analysis** showing that short moments (<10s) are the dominant failure mode, with models over-predicting moment length by 7× for this category.
- **Verb vs. noun sensitivity ablation** demonstrating that object nouns contribute 2.6× more to retrieval performance than action verbs.
- **Attention head ablation** revealing that temporal grounding is a distributed emergent property—no single attention head is specialized for boundary prediction.

2 Related Work

Video Moment Retrieval. Gao et al. [2017] introduced the temporal sentence grounding task and proposed TALL, a proposal-based approach using sliding windows. Subsequent work explored span-based methods [Zhang et al., 2020] and proposal-free approaches. Lei et al. [2021] proposed Moment-DETR, adapting the DETR framework [Carion et al., 2020] for VMR by treating moment prediction as set prediction with learned queries. This eliminated the need for hand-crafted proposals and enabled end-to-end training.

Query-Aware Models. QD-DETR [Moon et al., 2023b] improved upon Moment-DETR by making video representations query-dependent, enabling better discrimination between similar events. CG-DETR [Moon et al., 2023a] further introduced calibrated cross-attention that weights word-to-moment relevance, achieving the strongest fine-grained alignment among DETR-style models.

Bias and Interpretability in VMR. Lan et al. [2023] examined temporal biases in sentence grounding datasets, finding that models can exploit distributional shortcuts. Our work extends this analysis to DETR-style architectures with a focus on mechanistic interpretability through attention head ablation, a technique inspired by recent work in language model interpretability.

3 Methodology

3.1 Dataset

We use the QVHighlights benchmark [Lei et al., 2021], which contains over 10,000 natural language queries paired with videos of approximately 150 seconds. Each query is annotated with temporal moment boundaries and per-clip saliency scores for highlight detection. We evaluate on the standard validation split.

3.2 Feature Extraction

Following the standard protocol, we use pre-extracted multimodal features:

- **Visual:** CLIP ViT-B/32 [Radford et al., 2021] + SlowFast [Feichtenhofer et al., 2019] features concatenated along the feature dimension.
- **Text:** CLIP ViT-B/32 text encoder [Radford et al., 2021], producing both pooled and per-token representations.
- **Audio:** PANN [Kong et al., 2020] features (used by some configurations).

All features are packed into HDF5 format for efficient I/O during training on the Boston University Shared Computing Cluster (SCC).

3.3 Baseline Models

We train and evaluate three DETR-style models using the Lighthouse framework [Nishimura et al., 2024]:

Moment-DETR. Moment-DETR [Lei et al., 2021] uses a transformer encoder-decoder architecture with learnable moment queries. The encoder processes concatenated video and text features, while the decoder cross-attends to encoder outputs to predict moment boundaries via set prediction with bipartite matching loss.

QD-DETR. QD-DETR [Moon et al., 2023b] extends Moment-DETR by conditioning video representations on the text query through a query-dependent attention mechanism. This makes the video encoding query-aware, improving discrimination between visually similar but semantically distinct moments.

CG-DETR. CG-DETR [Moon et al., 2023a] further introduces calibrated grounding that explicitly models word-to-moment relevance. By weighting the contribution of individual words during cross-modal alignment, CG-DETR achieves the finest-grained grounding among the three models.

3.4 Evaluation Metrics

We report the following standard metrics:

- **R1@0.5, R1@0.7:** Recall@1 at IoU thresholds 0.5 and 0.7. The fraction of queries where the

top-1 predicted moment has IoU $\geq$ threshold with the ground truth.
- **mAP:** Mean average precision over IoU thresholds.
- **HL-mAP:** Highlight detection mAP under "Fair" and "Good" saliency standards.

3.5 Analysis 1: Temporal Bias

To assess whether models exploit temporal shortcuts, we compare model performance against three trivial baselines:

1. **Random guess:** Uniformly sample random $[t_s, t_e]$ intervals.
2. **Center-of-video:** Always predict a fixed-length segment centered at the video midpoint.
3. **Training distribution:** Sample moments from the empirical training set distribution of start/end times.

We additionally bucket predictions by ground-truth moment length (short: <10s, medium: 10–30s, long: 30–150s) and temporal position (early: first third, middle: second third, late: final third) to identify systematic failure modes.

3.6 Analysis 2: Verb vs. Noun Sensitivity

To understand which linguistic components drive grounding, we perform a part-of-speech (POS) ablation:

1. Use spaCy [Honnibal et al., 2020] to POS-tag all validation queries, identifying verbs and nouns.
2. Use the CLIP tokenizer to map word-level POS tags to subword token positions in the CLIP text representation.
3. Create two ablated feature sets by zeroing out either all verb token features or all noun token features in the CLIP `last_hidden_state`.
4. Re-evaluate QD-DETR with each ablated feature set and compare the R1@0.5 degradation.

The alignment between spaCy word-level tags and CLIP subword tokens requires careful handling: we tokenize each word individually to identify its subword IDs, then walk through the full-sentence token sequence to find matching positions (accounting for the start-of-sequence offset).

3.7 Analysis 3: Attention Head Ablation

Inspired by prior attention-head analysis and pruning work in Transformer models [Voita et al., 2019, Michel et al., 2019], we investigate whether specific attention heads are specialized for temporal boundary prediction.

QD-DETR contains 4 multi-head attention modules (2 text-to-video encoder layers + 2 main encoder layers), each with 8 heads, totaling 32 attention heads. For each head, we:

1. Register a forward hook that zeros out the output dimensions corresponding to that head.
2. Run full evaluation on the validation set.
3. Record the change in R1@0.5 relative to the unmodified model.

If a head is specialized for boundary prediction, zeroing it should cause a large performance drop. If grounding is distributed, all drops should be small.

Table 1: Baseline performance on QVHighlights validation set. All models use CLIP+SlowFast visual features.

Model	R1@0.5	R1@0.7	mAP	HL-F	HL-G
Moment-DETR	54.06	36.52	32.80	68.18	58.29
QD-DETR	62.06	45.48	40.56	74.14	63.30
CG-DETR	**62.84**	**46.84**	**42.00**	**76.05**	**64.88**

Table 2: Comparison with trivial bias baselines.

Method	R1@0.5
Random guess	8.89
Center of video	8.65
Training distribution	8.26
QD-DETR	**62.06**

4 Experiments and Results

4.1 Baseline Reproduction

Table 1 shows the performance of the three reproduced baselines on the QVHighlights validation set.

QD-DETR achieves a substantial +8.0 R1@0.5 improvement over Moment-DETR, demonstrating the value of query-dependent video representations. CG-DETR provides a further modest improvement through calibrated cross-attention, achieving the highest scores across all metrics.

4.2 Temporal Bias Analysis

Bias baselines. Table 2 compares model performance against trivial baselines. All three baselines achieve R1@0.5 below 9.0, while QD-DETR reaches 62.06—a 7× improvement. This confirms that the models perform genuine cross-modal grounding rather than exploiting distributional shortcuts.

Table 3: Performance by ground-truth moment length for QD-DETR. Short moments are the dominant failure mode.

Length	R1@0.5	Mean pred	Mean GT
Short (<10s)	50.00	32.0s	4.6s
Medium (10–30s)	64.68	30.0s	19.2s
Long (30–150s)	62.98	56.9s	61.2s

Table 4: Effect of zeroing verb vs. noun token features on QD-DETR performance.

Condition	R1@.5	R1@.7	mAP	ΔR1
Original	61.29	44.19	39.93	—
Verb-masked	60.39	44.13	39.14	−0.90
Noun-masked	58.97	42.90	37.91	−2.32

We observe a mild center bias: the mean predicted moment center is at normalized position 0.49, compared to 0.43 for the ground truth. However, this bias is modest and does not explain the models' strong performance.

Length-bucketed analysis. Table 3 reveals a striking failure pattern. Short moments (<10s) suffer roughly 12–16 point R1@0.5 drops across all models. Critically, QD-DETR predicts a mean length of 32.0s for short moments whose actual mean length is only 4.6s—an over-prediction of nearly 7×.

This suggests that the models have learned a prior toward medium-length predictions and struggle to produce the tight boundaries required for short moments. The length over-prediction is the primary driver of failures on short queries.

4.3 Verb vs. Noun Sensitivity

Across the validation set, queries contain on average 1.33 verbs and 3.75 nouns per query (2,063 verb-tagged words and 5,805 noun/proper-noun words total). During feature preparation, this corresponded to zeroing 12,405 verb subword positions and 39,548 noun/proper-noun subword positions across the train and validation feature files.

Table 4 shows that masking nouns causes a 2.6× larger R1@0.5 drop than masking verbs (−2.32 vs. −0.90). This effect is consistent across R1@0.7 (−1.29 vs. −0.06) and mAP (−2.02 vs. −0.79).

This finding indicates that QD-DETR grounds moments primarily through entity and object recognition rather than action semantics. While the model can identify *what* objects are present, it is relatively insensitive to *what is being done* with them. This represents a potential vulnerability for action-heavy queries where the distinguishing factor between moments is the verb (e.g., "picks up the cup" vs. "puts down the cup").

Table 5: Top-5 most impactful attention heads when individually ablated. Maximum drop is only −0.77 R1@0.5 points from a baseline of 61.29.

Head	ΔR1@0.5
`encoder.layer0.head3`	−0.77
`t2v_encoder.layer1.head2`	−0.39
`encoder.layer0.head1`	−0.32
`encoder.layer1.head3`	−0.32
`t2v_encoder.layer0.head0`	−0.13

4.4 Attention Head Ablation

Table 5 shows the top 5 most impactful heads when individually zeroed.

The key finding is that the maximum single-head drop is only −0.77 R1@0.5 points (from a baseline of 61.29), representing just a 1.3% relative decrease. Only 4 of 32 heads reduce R1@0.5 by at least 0.3 points when removed, and several ablations slightly improve R1@0.5, indicating that no single head is critical on its own.

Two patterns emerge from the ablation:

1. **Ablation effects are small and uneven.** The largest negative effects appear in both main encoder and text-to-video encoder layers, suggesting that individual heads provide only weak, non-localized contributions to grounding.
2. **Temporal grounding is distributed.** No single head acts as a "boundary detector." The model's ability to predict temporal boundaries is an emergent property of the collective computation across all 32 heads, not localized in any specific component.

This distributed representation makes the model robust to individual head failures but also means that targeted interventions to improve specific aspects of grounding (e.g., boundary precision) cannot simply modify a single component.

5 Discussion

Our three analyses provide complementary perspectives on how DETR-style VMR models work and fail.

Short moments as a design blind spot. The temporal bias analysis reveals that the dominant failure mode is not distributional bias but rather an inability to produce tight temporal boundaries. Models have learned a prior toward medium-length predictions (approximately 30s), which works well for the most common moment lengths but catastrophically overpredicts for short moments. This likely stems from the training objective: L1 and IoU losses penalize absolute boundary errors, and the absolute error required to miss a 5-second moment at IoU 0.5 is much smaller than for a 50-second moment.

Object-centric grounding. The 2.6× greater importance of nouns over verbs suggests that these models primarily perform a form of "object spotting"—they identify when queried objects appear in the video rather than understanding the actions being performed. This is consistent with CLIP's known strength at object recognition and relative weakness at action understanding. For datasets where queries primarily differ in which objects are mentioned, this strategy is effective. However, it represents a fundamental limitation for queries that require fine-grained action discrimination.

Distributed computation. The head ablation finding that no single head is specialized for boundary prediction aligns with prior Transformer head-pruning studies, where many heads can be removed with limited performance loss [Voita et al., 2019, Michel et al., 2019]. This suggests that attempts to improve VMR through targeted architectural modifications may need to consider the full network rather than individual components.

Implications for future architectures. These findings collectively suggest three design directions: (1) loss functions or training strategies that explicitly handle short moments, (2) video encoders with stronger action semantics beyond CLIP, and (3) architectures that allow more structured decomposition of the grounding task to enable targeted improvements.

6 Conclusion and Future Work

We presented an interpretability and failure analysis of three DETR-style video moment retrieval models on the QVHighlights benchmark. Our key findings are:

1. Models achieve 7× above trivial baselines, confirming genuine cross-modal grounding, but systematically fail on short moments (<10s) due to length over-prediction.
2. Object nouns are 2.6× more important than action verbs for QD-DETR's grounding decisions, revealing an object-centric bias.
3. Temporal boundary prediction is a distributed emergent property across all 32 attention heads, with no specialized boundary heads.

Future work includes: (1) multi-head ablation testing combinations of heads rather than individuals; (2) activation patching to trace causal information flow; (3) short-moment augmentation strategies during training; (4) cross-dataset analysis on Charades-STA to test generality; and (5) incorporating action-specialized video-language models to address the verb sensitivity gap.

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